

The Complete Guide to Shavian, Tengwar, and Featural Transliteration

Part 1: The Shavian Alphabet & ReadLex Specification

The Shavian alphabet is a purely phonemic writing system designed by Kingsley Read to fulfill the bequest of George Bernard Shaw. It replaces traditional non-phonetic English spelling with a 1:1 sound-to-letter mapping.

1. Character Taxonomy

Shavian characters fall into four structural categories based on vertical height and composition:

- **Tall (Voiceless Consonants):** Extend above the x-height.
- **Deep (Voiced Consonants):** Extend below the baseline.
- **Short (Vowels, Liquids, Nasals):** Fit entirely within the x-height.
- **Compound (Vocalic Ligatures):** Combine rhotic (/r/) sounds and vowel clusters into single glyphs.

Consonant Sound Pairs

Tall and Deep consonants form physical pairs where rotating or flipping a Tall (voiceless) letter generates its Deep (voiced) counterpart.

Sound Pair	Tall Glyph (Voiceless)	Deep Glyph (Voiced)	Phonetic Names
/p/ – /b/	ᵛ	ᵇ	peep – bib
/t/ – /d/	ᵀ	ᵀ̣	tot – dead
/k/ – /g/	ᵀ̣	ᵇ	kick – gag
/f/ – /v/	J	ᵀ̣	fee – vow
/θ/ – /ð/	ᵀ̣	ᵀ̣̣	thigh – they
/s/ – /z/	S	ᵀ̣̣	so – zoo

Sound Pair	Tall Glyph (Voiceless)	Deep Glyph (Voiced)	Phonetic Names
/ʃ/ – /ʒ/	ℓ	ɹ	sure – measure
/tʃ/ – /dʒ/	ℓ	ɹ	church – judge
/h/ – /w/	ɸ	/	he – woe

Vowels, Liquids, and Compound Ligatures

Category	Characters & Keywords
Liquids & Nasals	ℓ (/l/ loll), ʃ (/m/ mime), ʌ (/n/ nun), ɹ (/ŋ/ hung), ɔ (/r/ roar), ʌ (/j/ yea)
Short Vowels	ɪ (/ɪ/ if), ʌ (/ɛ/ egg), ʌ (/æ/ ash), ɹ (/ə/ ado), ɹ (/ʌ/ up), ʌ (/ɒ/ on), ʌ (/ʊ/ wool)
Long Vowels	ɪ (/i:/ eat), ɹ (/eɪ/ age), ɹ (/ɑ:/ ah), ɹ (/ɔ:/ awe), ʌ (/oʊ/ oak), ʌ (/u:/ ooze)
Diphthongs	ɹ (/aɪ/ ice), ɹ (/ɔɪ/ oil), ɹ (/aʊ/ out)
Compound Ligatures	ɹ (/ɑ:r/ are), ɹ (/ɔ:r/ or), ɹ (/ɛər/ air), ɹ (/ɜ:r/ err), ɹ (/ər/ array), ɹ (/ɪər/ ear), ɹ (/iə/ Ian), ʌ (/ju:/ yew)

2. The 12 ReadLex Orthographic Rules

- Strict Sound-to-Letter Phonetics:** Write strictly by spoken sound, ignoring silent Latin letters (e.g., *write* = ɹɹɪ).
- Phonetic Voicing Assimilation in Suffixes:** Suffixes match spoken voicing (e.g., *cats* = ɹʌʃ vs. *dogs* = ɹɹɹ; *walked* = ɹɹɹ vs. *played* = ɹɹɹ).
- The Big Five Abbreviations:** Five high-frequency function words are abbreviated as single shorthand letters: *the* = ʌ, *of* = ʃ, *and* = ʌ, *to* = 1, *for* = J.
- Monosyllabic Invariance:** Single-syllable words maintain their fully stressed spelling regardless of context (e.g., *but* is always ɹɹɹ).
- Monosyllabic Weak Vowel Exception:** Indefinite articles are explicitly written weak using schwa (ɹ): *a* = ɹ, *an* = ɹ.

6. **Stress Distinction in Polysyllabic Words:** Vowel reduction reflects stress shifts (e.g., noun *convict* = ɔ̃ʌʋɪɔ̃ vs. verb *convict* = ɔ̃ʌʋɪɔ̃).
 7. **No Letter Gemination:** Consonants are never doubled unless articulated twice across syllable boundaries (e.g., *annoy* = ʌʌ vs. *unnamed* = ʌʌʌʌ).
 8. **Syllabic Consonant Formatting:** Insert a schwa (ɪ) before terminal liquids or nasals (e.g., *little* = ɔ̃ʌʌɪ, *driven* = ɔ̃ʌʌʌ).
 9. **Verbal Contraction Formatting:** Negative contractions insert ɪ before ʌ and omit apostrophes (e.g., *didn't* = ɔ̃ʌʌʌ).
 10. **Proper Noun Marking:** Capital letters are omitted entirely; proper nouns use a middle Namer Dot (·) immediately preceding the word (e.g., *London* = ·ɔ̃ʌʌʌ).
 11. **Unstressed Terminal Vowels (happy Rule):** Words ending in unstressed -y or -ie use ɪ (e.g., *happy* = ʌʌʌ). Tense, fully stressed final vowels use ɪ (e.g., *trustee* = ʌʌʌʌ).
 12. **Cross-Dialectal "Rhotic RP" Standard:** Standard spelling uses a broad rhotic Received Pronunciation base so rhotic and non-rhotic speakers share a single format.
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Part 2: Tolkien's Tengwar & CSUR Standard

Tengwar is a featural abugida created by J.R.R. Tolkien. Primary consonants (*tengwar*) represent structural articulation features, while vowels (*tehtar*) sit above or below consonant bases as diacritics.

1. The Fëanorian Consonant Grid (Appendix E Layout)

In Tolkien's standard layout, primary consonants are organized into **4 Series** (Columns: Place of Articulation) and **6 Grades** (Rows: Manner of Articulation and Voicing):

- **Series I (Témar 1):** Dental / Alveolar (/t/, /d/, /θ/, /ð/)
- **Series II (Témar 2):** Labial / Bilabial (/p/, /b/, /f/, /v/)
- **Series III (Témar 3):** Palatal / Affricate (/tʃ/, /dʒ/, /ʃ/, /ʒ/)
- **Series IV (Témar 4):** Velar (/k/, /g/, /ŋ/)

Grade / Manner	Series I (Dental)	Series II (Labial)	Series III (Palatal)	Series IV (Velar)
Grade 1: Voiceless Stops (<i>Stem Down, Single Loop</i>)	p U+E000 <i>Tinco</i> (/t/)	p U+E001 <i>Parma</i> (/p/)	ç U+E002 <i>Calma</i> (/tʃ/)	q U+E003 <i>Quesse</i> (/k/)
Grade 2: Voiced Stops (<i>Stem Down, Double Loop</i>)	ɖ U+E004 <i>Ando</i> (/d/)	ɸ U+E005 <i>Umbar</i> (/b/)	çç U+E006 <i>Anga</i> (/dʒ/)	ɕ U+E007 <i>Ungwe</i> (/g/)
Grade 3: Voiceless Fricatives (<i>Stem Up, Single Loop</i>)	θ U+E008 <i>Thúle</i> (/θ/)	ɸ U+E009 <i>Formen</i> (/f/)	ɕ U+E00A <i>Harma</i> (/ʃ/)	ɕ U+E00B <i>Hwesta</i> (/hw/)
Grade 4: Voiced Fricatives (<i>Stem Up, Double Loop</i>)	ð U+E00C <i>Anta</i> (/ð/)	ɸ U+E00D <i>Ampa</i> (/v/)	çç U+E00E <i>Anca</i> (/ʒ/)	ɕ U+E00F <i>Unque</i> (/gh/)
Grade 5: Nasals (<i>Short Stem, Double Loop</i>)	ɳ U+E010 <i>Númen</i> (/n/)	ɹ U+E011 <i>Malta</i> (/m/)	çç U+E012 <i>Noldo</i> (/ŋ/)	ɕ U+E013 <i>Nwalme</i> (/nw/)
Grade 6: Glides / Semi-vowels (<i>Short Stem, Single Loop</i>)	ɹ U+E014 <i>Óre</i> (/r/ final)	ɹ U+E015 <i>Vala</i> (/w/)	ɕ U+E016 <i>Anna</i> (/j/)	ɕ U+E017 <i>Vilya</i>

2. Additional & Irregular Tengwar

Beyond the standard 24-tengwa grid, specialized base characters handle liquids, sibilants, glottal sounds, and silent carriers:

Tengwa Glyph	Code Point	Tengwa Name	Structural Description / Function
ṛ	U+E018	<i>Rómen</i>	Trilled/vocalic /r/ (used before vowels)
ḷ	U+E01A	<i>Lamba</i>	Upward hooked liquid character (/l/)
ś	U+E024	<i>Silme</i>	Upright S-curve sibilant (/s/)
ṣ	U+E025	<i>Silme Nuquerna</i>	Inverted S-curve (used when carrying top tehta)
z	U+E026	<i>Esse / Aze</i>	Z-curve / inverted double loop (/z/)
ḥ	U+E027	<i>Esse Nuquerna</i>	Inverted Z-curve (used when carrying top tehta)
h	U+E020	<i>Hyarmen</i>	Downward stem with left hook (/h/)
ˆ	U+E028	<i>Telco</i> (Short Carrier)	Unadorned vertical pillar carrying short orphan vowels
˙	U+E029	<i>Ára</i> (Long Carrier)	Extended vertical pillar carrying long orphan vowels

3. Vowel Tehtar & Diacritic Marks

In English General Mode, vowel diacritics (*tehtar*) sit directly above or below the following consonant base glyph:

Tehta Mark	CSUR Hex	Tehta Name	Phonetic Mapping	Visual Placement
ˆ	U+E040	<i>a-tehta</i>	/æ/, /ɑː/, /aɪ/	Three dots in a triangle above base
˙	U+E044	<i>i-tehta</i>	/ɪ/, /iː/	Single dot above base
˘	U+E045	<i>schwa-tehta</i>	/ə/ (schwa)	Single dot below base

Tehta Mark	CSUR Hex	Tehta Name	Phonetic Mapping	Visual Placement
ˊ	U+E046	<i>e-tehta</i>	/ɛ/, /eɪ/	Single acute stroke above base
ˊˊ	U+E047	<i>long e-tehta</i>	/iː/ (FLEECE)	Double acute stroke above base
ˆ	U+E04A	<i>o-tehta</i>	/ɒ/, /ɔː/, /oʊ/	Right-facing curl above base
ˋ	U+E04C	<i>u-tehta</i>	/ʌ/, /ʊ/, /uː/	Left-facing curl above base
ˋˋ	U+E04D	<i>long u-tehta</i>	/uː/ (GOOSE)	Double left-facing curl above base
◌̃	U+E04E	<i>nasal bar</i>	Preconsonantal nasal	Horizontal tilde above base stop

Part 3: The Shavian-to-Tengwar Transliteration Bridge

By using Shavian as an Intermediate Representation (IR), transliteration bypasses non-phonetic English orthography entirely, converting text through a 5-pass state machine.

1. Direct Phoneme Mapping Matrix

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	CSUR Hex Stream	State Machine Transformation Rule
/p/	ᑭ	ᑭ	<i>Parma</i>	U+E001	Direct 1:1 Consonant Base
/b/	ᑭ̃	ᑭ̃	<i>Umbar</i>	U+E005	Direct 1:1 Consonant Base
/t/	ᑭ̌	ᑭ̌	<i>Tinco</i>	U+E000	Direct 1:1 Consonant Base
/d/	ᑭ̎	ᑭ̎	<i>Ando</i>	U+E004	Direct 1:1 Consonant Base
/k/	ᑭ̇	ᑭ̇	<i>Quesse</i>	U+E003	Direct 1:1 Consonant Base

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	CSUR Hex Stream	State Machine Transformation Rule
/g/	᠑	᠑	<i>Ungwe</i>	U+E007	Direct 1:1 Consonant Base
/f/	J	᠒	<i>Formen</i>	U+E009	Direct 1:1 Consonant Base
/v/	᠒	᠓	<i>Ampa</i>	U+E00D	Direct 1:1 Consonant Base
/θ/	᠓	᠔	<i>Thúle</i>	U+E008	Direct 1:1 Consonant Base
/ð/	᠔	᠕	<i>Anta</i>	U+E00C	Direct 1:1 Consonant Base
/s/	S	᠖	<i>Silme / Nuquerna</i>	U+E024 / U+E025	Flips to <i>Silme Nuquerna</i> (U+E025) if top tehta present
/z/	᠖	᠗	<i>Esse / Nuquerna</i>	U+E026 / U+E027	Flips to <i>Esse Nuquerna</i> (U+E027) if top tehta present
/ʃ/	᠗	᠘	<i>Harma</i>	U+E00A	Direct 1:1 Consonant Base
/ʒ/	᠘	᠙	<i>Anca</i>	U+E00E	Direct 1:1 Consonant Base
/tʃ/	᠙	᠐	<i>Calma</i>	U+E002	Direct 1:1 Consonant Base
/dʒ/	᠐	᠑	<i>Anga</i>	U+E006	Direct 1:1 Consonant Base
/j/	᠑	᠒	<i>Anna</i>	U+E016	Direct 1:1 Consonant Base
/w/	᠒	᠓	<i>Vala</i>	U+E015	Direct 1:1 Consonant Base
/ŋ/	᠓	᠔	<i>Noldo</i>	U+E012	Direct 1:1 Consonant Base
/h/	᠔	᠕	<i>Hyarmen</i>	U+E020	Direct 1:1 Consonant Base
/l/	᠕	᠖	<i>Lamba</i>	U+E01A	Direct 1:1 Consonant Base
/m/	᠖	᠗	<i>Malta</i>	U+E011	Direct 1:1 Consonant Base

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	CSUR Hex Stream	State Machine Transformation Rule
/n/	᳚	᳚	<i>Númen</i>	U+E010	Direct 1:1 Consonant Base
/r/	᳚	᳚ / ᳚	<i>Rómen / Óre</i>	U+E018 / U+E014	Contextual: <i>Rómen</i> before vowels; <i>Óre</i> word-final/consonantal
/ɪ/	᳚	᳚	<i>i-tehta</i>	U+E044	Attach dot above next available consonant
/i:/	᳚	᳚	<i>Ára + i-tehta</i>	U+E029 U+E044	FLEECE vowel: Attach to Long Carrier or double acute on base
/ɛ/	᳚	᳚	<i>e-tehta</i>	U+E046	Attach acute stroke above next available consonant
/eɪ/	᳚	᳚	<i>Anna + e-tehta</i>	U+E016 U+E046	FACE diphthong: Offglide <i>Anna</i> base + <i>e-tehta</i>
/æ/	᳚	᳚	<i>a-tehta</i>	U+E040	Attach three dots above next available consonant
/aɪ/	᳚	᳚	<i>Anna + a-tehta</i>	U+E016 U+E040	PRICE diphthong: Offglide <i>Anna</i> base + <i>a-tehta</i>
/ə/	᳚	᳚	<i>schwa-tehta</i>	U+E045	Attach single dot below next available consonant
/ʌ/	᳚	᳚	<i>u-tehta</i>	U+E04C	STRUT vowel: Attach left curl above next available consonant
/ɒ/	᳚	᳚	<i>o-tehta</i>	U+E04A	LOT vowel: Attach right curl above next available consonant
/oʊ/	᳚	᳚	<i>Vala + o-tehta</i>	U+E015 U+E04A	GOAT diphthong: Offglide <i>Vala</i> base

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	CSUR Hex Stream	State Machine Transformation Rule
					+ <i>o-tehta</i>
/ʊ/	ᵛ	ᚦ	<i>foot-tehta</i>	U+E04C	FOOT vowel: Attach left curl above next available consonant
/uː/	ᵛ	ᚦ	<i>Ára + u-tehta</i>	U+E029 U+E04C	GOOSE vowel: Attach to Long Carrier or double curl on base
/aʊ/	ᵛ	ᚦ	<i>Vala + a-tehta</i>	U+E015 U+E040	MOUTH diphthong: Offglide <i>Vala</i> base + <i>a-tehta</i>
/ɔɪ/	ᵛ	ᚦ	<i>Anna + o-tehta</i>	U+E016 U+E04A	CHOICE diphthong: Offglide <i>Anna</i> base + <i>o-tehta</i>
/ɑː/	ᵛ	ᚦ	<i>a-tehta</i>	U+E040	PALM vowel: Attach three dots above next available consonant
/ɔː/	ᵛ	ᚦ	<i>awe-tehta</i>	U+E04A	THOUGHT vowel: Attach right curl above next available consonant
/iə/	ᵛ	ᚦ	<i>Anna + i-tehta</i>	U+E016 U+E044	IAN ligature: Offglide <i>Anna</i> base + <i>i-tehta</i>
/juː/	ᵛ	ᚦ	<i>Anna + u-tehta</i>	U+E016 U+E04C	YEW ligature: Offglide <i>Anna</i> base + <i>u-tehta</i>
/ɑɪr/	ᵛ	ᚦ	<i>Rómen + a-tehta</i>	U+E018 U+E040	START vowel: Decomposes into <i>Rómen</i> + <i>a-tehta</i>
/ɔɪr/	ᵛ	ᚦ	<i>Rómen + o-tehta</i>	U+E018 U+E04A	NORTH vowel: Decomposes into <i>Rómen</i> + <i>o-tehta</i>
/ər/	ᵛ	ᚦ	<i>Óre</i>	U+E014	letter vowel: Decomposes into unadorned <i>Óre</i>

IPA Phoneme	Shavian Glyph	Tengwa	Target Tengwa / Tehta	CSUR Hex Stream	State Machine Transformation Rule
/ɪər/	Ꭰ	þ	<i>Rómen + i-tehta</i>	U+E018 U+E044	NEAR vowel: Decomposes into <i>Rómen + i-tehta</i>
/ɛər/	Ꭱ	þ	<i>Rómen + e-tehta</i>	U+E018 U+E046	SQUARE vowel: Decomposes into <i>Rómen + e-tehta</i>
/ʊr/	Ꭲ	þ	<i>Rómen + u-tehta</i>	U+E018 U+E04C	NURSE vowel: Decomposes into <i>Rómen + u-tehta</i>

2. Verified Test Cases & Output Comparison

To verify visual output in Obsidian or PDF export, install [Alcarin Tengwar](#). The table below lists the exact Shavian input, rendered Tengwar output, CSUR hex bytes, and feature rules applied:

Word	Shavian Input	CSUR Tengwar Output	CSUR Hex Stream	Applied Rule / Logic
winter	ᎠᎠᎠᎡ	ᎠᎠᎠᎡᎢ	U+E015 U+E010 U+E044 U+E000 U+E014	<i>i-tehta</i> attaches to <i>Númen</i> ; <i>Tinco</i> follows; ends in <i>Óre</i> .
chamber	ᎡᎢᎢᎡ	ᎡᎢᎢᎡᎢ	U+E002 U+E016 U+E046 U+E005 U+E04E U+E014	Nasal pair ᎢᎢ lacks vowel, emitting <i>Umbar + Nasal Bar Above</i> (U+E04E).
third	ᎢᎢᎢ	ᎢᎢᎢᎢ	U+E008 U+E018 U+E04C U+E004	NURSE vowel Ꭲ decomposes into <i>Rómen + u-tehta</i> (U+E04C).
hair	ᎢᎢ	ᎢᎢ	U+E020 U+E018 U+E046	SQUARE vowel Ꭲ decomposes into <i>Rómen + e-tehta</i> (U+E046).
bruised	ᎢᎢᎢᎢ	ᎢᎢᎢᎢᎢ	U+E005 U+E018 U+E01F U+E04D	Carrying double <i>u-curl</i> forces <i>Esse</i> to

Word	Shavian Input	CSUR Tengwar Output	CSUR Hex Stream	Applied Rule / Logic
			U+E004	flip to <i>Esse Nuquerna</i> (U+E01F).
merry	ꞑꞐꞑꞓ	ṡþλ	U+E011 U+E018 U+E046 U+E028 U+E044	Word-final short vowel attaches to <i>Short Carrier</i> (<i>Telco</i> , U+E028).
he	ꝥꝨ	ꝥð	U+E020 U+E029 U+E044	Word-final long vowel attaches to <i>Long Carrier</i> (<i>Ára</i> , U+E029).
lady's	ꝥꝥꝒꝑ'Ꝛ	ꝥꝥꝖꝗꝕꝔ'ꝥ	U+E01A U+E016 U+E046 U+E004 U+E028 U+E044 U+2019 U+E01E	Contraction apostrophe splits token; trailing <i>Esse</i> is unadorned.